

Part 2

2.2.1 Followup questions

1. Performance Comparison:

Setting	I2T R@1	I2T R@5	I2T R@10	T2I R@1	T2I R@5	T2I R@10
Zero-shot Baseline	0.4653	0.7303	0.8183	0.4558	0.7044	0.8101
InfoNCE (1e-5, E3)	0.4538	0.7293	0.8221	0.4720	0.7465	0.8374
Triplet (1e-5, E3)	0.4596	0.7331	0.8274	0.4586	0.7303	0.8168

After fine-tuning, both losses stay close to the zero-shot baseline. InfoNCE lifts the text-to-image recall the most, especially the R@10 score of 0.8374. Triplet keeps a small edge on image-to-text R@5 and R@10.

The gap is narrow because we only update the projection heads and the dataset is modest in size. For this dataset, InfoNCE performs better as it shows a more significant performance uplift, particularly on Text-to-Image retrieval tasks. InfoNCE considers all positive and negative pairs in the batch, so the gradient signal stays steady. Triplet uses one rolled negative per anchor, so the negative can be too easy and the loss stops improving early. In this case, InfoNCE might be better choice for extending tuning.

2. Training Loss Analysis:

We trained two models on Flickr8k using different objectives: InfoNCE and Triplet loss. For both experiments, we logged the training loss per epoch using SummaryWriter under the runs/ directory and visualised the curves with TensorBoard via:

```
tensorboard --logdir runs
```

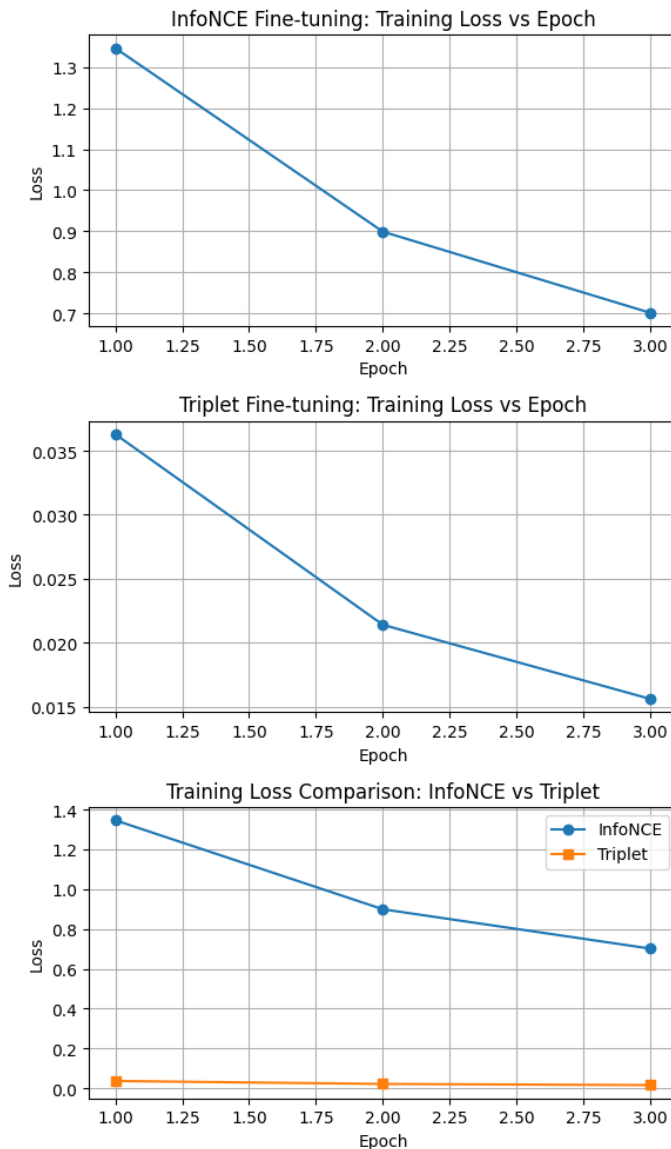
For the sake of convenience, we paste images generated by matplotlib here in the report.(next page)

Loss Analysis:

InfoNCE loss drops sharply (1.346 → 0.701) over three epochs, indicating the contrastive objective quickly stabilizes; retrieval metrics barely move (I2T_R@1 stays ~0.46–0.47, T2I_R@1 ~0.45–0.47), so the rapid loss decrease is not translating into noticeable ranking gains—suggesting the model may already be near a local optimum or overfitting to the loss without improving ranking.

Triplet loss begins much lower (0.036) and keeps shrinking (0.021 → 0.016); despite the lower scale, recall improves slightly for image-to-text (I2T_R@1 inching up to 0.4735) but text-to-image declines a bit by epoch 3 (T2I_R@1 0.458 → 0.451), implying the margin-based fine-tuning is giving marginal I2T gains while hurting T2I after epoch 2.

Comparing the two, infoNCE delivers balanced recall between directions and decreasing steadily, whereas Triplet loss converges faster in terms of epochs. In other words, Triplet training is “fast but short”, whereas InfoNCE training is “slower but progressive” (smooth) over a larger loss range.



3. Hyperparameter Experimentation (Learning Rate):

In the baseline setting, we trained the projection layers using a learning rate in the middle range. To understand how sensitive the model is to this choice, we ran the same training procedure again with one rate that was clearly higher and another that was much lower.

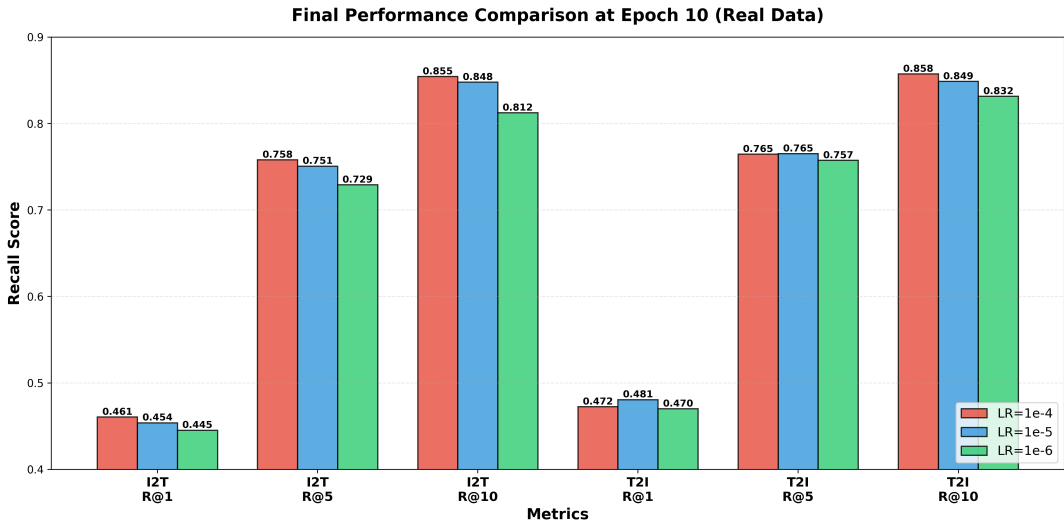
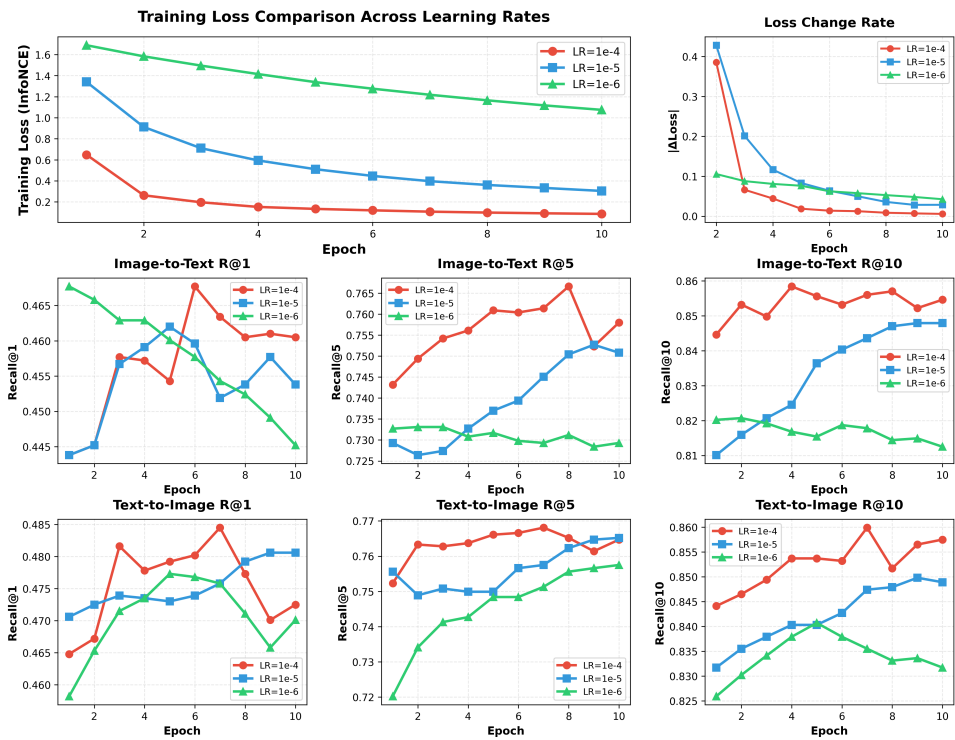
We chose 10 epochs because baseline results showed that learning saturates by epoch 10, with both loss and retrieval metrics stabilizing. This keeps training efficient while ensuring fair comparisons.

With the higher learning rate, the loss dropped quickly but the training became unstable. The curve fluctuated instead of moving smoothly, and the retrieval scores reflected this instability. For example, text-to-image Recall@1 improved early on but later slipped from around 0.48 to about 0.47, even though the training loss was the lowest among all runs. This suggests mild overfitting: the model learns fast but does not generalise well.

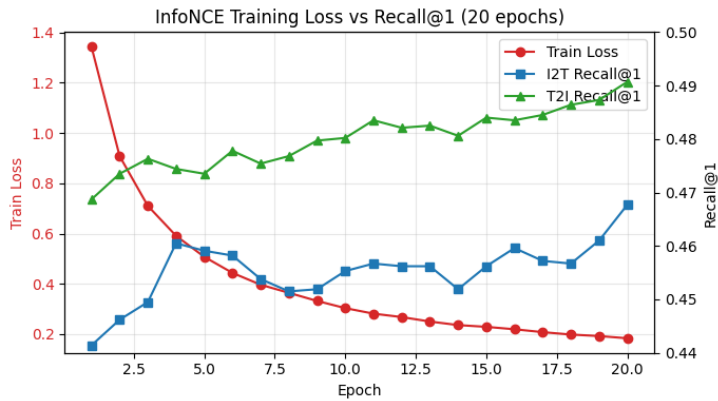
With the lower learning rate, the model barely progressed. Even after ten epochs, the loss stayed high and the recall metrics were the weakest. The steps were simply too small for the projection layers to adapt, so the model remained close to its zero-shot behaviour.

Comparing the three settings, the baseline learning rate offered the most balanced and consistent results. The higher rate was fast but unstable, and the lower one was too slow to be practical. The baseline remained the most reliable choice for this fine-tuning setup.

Hyperparameter Experimentation: Learning Rate Analysis



4. Overfitting Analysis:



There are mild signs of overfitting, but the impact is asymmetric across tasks.

The training loss keeps falling ($1.34 \rightarrow 0.18$), which signals stronger memorisation.

The validation curves split after epoch 4: image-to-text is volatile, peaking at 0.4605 in epoch 4, dipping to 0.4515 by epoch 8, and only recovering later. Text-to-image stays much more stable and keeps rising from 0.4744 to 0.4907 over the same period. So the overfitting noise mainly hurts I2T temporarily, while T2I keeps benefiting from longer training.

The optimal number of training epochs is 20.

Epoch 4 looks efficient, but it would cut off both the I2T recovery (up to 0.4677) and the steady T2I gains ($0.4744 \rightarrow 0.4907$). The later checkpoint proves the model is still learning useful representations, especially for T2I, so training to epoch 20 is worth the compute.

5. Qualitative Analysis:

Best-performing checkpoint: InfoNCE fine-tune with T2I_R@10: 0.841.

- **Image→Text Success:**



ground_truth_caption: "A man wearing a black hat is walking through a busy city street ."

Top-3 captions retrieved:

"A man wearing a black hat is walking through a busy city street." (GT, cosine 0.435)

"A man and a woman in a silver coat are on a city sidewalk with their backs to the camera." (0.415)

"Two men in hats stand on a wide sidewalk." (0.414)

This is a success because the model precisely matched the context (man in hat, urban street) and the distractor captions are semantically close but lack direct agreement with all details.

- **Image→Text Failure:**



ground_truth_caption: "A black and white dog a black dog and an orange dog are in the grass "

Top-3 captions retrieved:

"One dog jumps up to catch a ball as another dog runs towards it ."(GT, cosine 0.419)

"Two black and white dogs run across a green field ."(0.418)

"one brown and white dog chasing a black and white dog through the grass"(GT, cosine 0.412)

The retrieved captions (top-3) mention dogs running or chasing but mis-handle color/count details. This points to two gaps:

- The model encodes strong coarse semantics (dogs, outdoors, action) but struggles with multi-attribute conjunctions. Because Flickr8k contains many dog-running scenes with similar contexts, the projection heads latch onto dominant cues (species + motion) and ignore secondary ones (exact coloration, number of dogs).

- The imposed rolled-negative triplet/InfoNCE training doesn't explicitly encourage attribute disentanglement; positives and negatives often share "dog in park" features, so the embedding may collapse such scenes together.

- **Text→Image Success:**

Querying text: "A girl in a pink jacket is going headfirst down a red slide,"

the model retrieves three visually consistent images.



Top-1 (also the ground truth) "A girl in a pink jacket is going headfirst down a red slide,"



Top-2 shows a child climbing inside a red tube slide (same object, different pose).



Top-3 is another child inside a glowing red slide tunnel.

All results keep the dominant attributes (child + red slide playground), showing the fine-tuned InfoNCE model can align this caption with multiple semantically similar photos, not just the exact match.

The model succeeds because:

- The caption combines multiple distinctive cues that co-occur reliably—"girl", "pink jacket", "red slide". In the dataset there are only a handful of playground shots with that color palette, so during InfoNCE training those examples become well separated from other captions.
- The negatives are diverse (parks, beaches, animals), forcing the projection heads to attend to the specific conjunction of "child + pink outerwear + bright red slide/tunnel".

- **Text→Image Failure:**

Querying text: "A black and white dog, a black dog and an orange dog are in the grass."



Ground-truth image

Ground-truth shows three differently colored dogs running in a park, but the retrieved top 3 images were:



Top-1 two black dogs tugging a toy with a tan dog near a house.



Top-2 three dogs (white/brown/black) running but with different composition and lighting.



Top-3 two dogs (white/black and brown) running, missing the third dog and orange coloring.

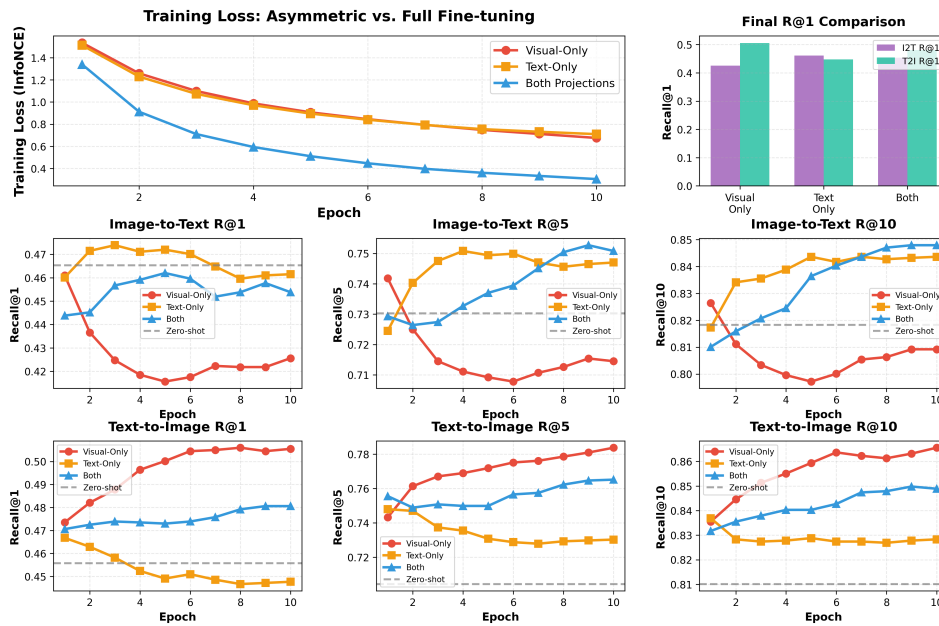
Likely failure causes:

The cues are less distinctive and more combinatorial. Flickr8k contains many “dogs running on grass” images that share the dominant features (species, outdoors, motion). The projection heads mostly learn to recognize that coarse concept. Color/count attributes (“black and white”, “orange”, “three dogs”) are harder to train.

6. Discussion Question (Bonus):

We investigated whether training only one projection layer (visual or text) could match the performance of training both. Three strategies were tested: Visual-Only (train visual projection only), Text-Only (train text projection only), and Both Projections (baseline). All used learning rate $1e-5$ for 10 epochs on Flickr8k.

Question 6: Asymmetric Fine-tuning Analysis



Visual-Only achieves the best Text-to-Image (T2I) performance: T2I R@1 of 0.5055 (5.2% better than Both), T2I R@5 of 0.7838 (2.4% better), and T2I R@10 of 0.8656 (2.0% better). Conversely, Text-Only excels at Image-to-Text (I2T) retrieval with I2T R@5 of 0.7470 and I2T R@10 of 0.8436, matching the Both baseline. However, Text-Only severely degrades T2I performance below the zero-shot baseline.

Asymmetric Fine-tuning Summary - Epoch 10

Setting	Final Loss	I2T R@1	I2T R@5	I2T R@10	T2I R@1	T2I R@5	T2I R@10
Zero-shot	N/A	0.4653	0.7303	0.8183	0.4558	0.7044	0.8101
Visual-Only	0.6757	0.4256	0.7145	0.8092	0.5055	0.7838	0.8656
Text-Only	0.7100	0.4615	0.7470	0.8436	0.4477	0.7303	0.8283
Both Projections	0.3042	0.4538	0.7508	0.8479	0.4806	0.7652	0.8489

From these outcomes, we infer that the visual side adapts more effectively on Flickr8k. The images show more variation from CLIP's original training distribution than the captions, so adjusting the visual projection seems to help more. In contrast, the text captions are short and descriptive, and CLIP already handles that style well. This may explain why updating only the text projection offers limited gains and can negatively affect tasks involving image retrieval.

Another interesting finding is that visual-only training outperforms training both projections. We infer two possible reasons:

1. Regularisation effect

Training only one projection layer reduces the chance of overfitting. The model that trained both projections reached the lowest training loss, but the retrieval performance did not reflect that. It likely fit the training set too closely without improving generalisation.

2. Uneven domain shift

Flickr8k images differ more from CLIP's pre-training images than its captions differ from CLIP's pre-training text. The text side might already be "good enough," while the visual side needs more adjustment. Updating both at once may cause the two projections to move in directions that don't complement each other perfectly.

Overall, the results show a clear asymmetry. For text-to-image retrieval, updating the visual projection is the most beneficial. For image-to-text retrieval, updating the text projection helps somewhat, but the improvement is not as pronounced.

In short, we infer that the visual projection is the main bottleneck for this dataset, and focusing on it gives the best balance between performance and efficiency.

AI Tool Declaration:

We used GPT-5 to generate ideas, improve expression, produce drafts, refine our assignment. We are responsible for the content and quality of the submitted work.